

A 5.5 GHz Double-Balanced Gilbert-Cell Down-Conversion Mixer in 180 nm RF CMOS

Design and Characterization Study

Min Aung Thu Hein Htut

Department of Robotics and Automation Engineering
Kasetsart University, Sriracha Campus, Chonburi, Thailand

Abstract—This paper presents a design and characterization study of a double-balanced Gilbert-cell down-conversion mixer, simulated at the schematic level in the TSMC 0.18 μm RF CMOS process. Operating from a 1.8 V supply, the mixer down-converts a 5.5 GHz RF input to a 200 MHz intermediate frequency (IF) using a 5.3 GHz local oscillator (LO). The mixer achieves a conversion gain of 17.9 dB, a single-sideband noise figure of 11.4 dB, an input 1-dB compression point of -19 dBm, and an input third-order intercept point (IIP3) of -9.9 dBm.

Index Terms—Gilbert cell, down-conversion mixer, double-balanced, CMOS, RF front-end, harmonic balance, conversion gain, IIP3.

I. INTRODUCTION

The down-conversion mixer is a critical block in an RF receiver front-end, translating the incoming RF signal to a lower intermediate frequency for subsequent baseband processing. Positioned after the low-noise amplifier (LNA) and before the IF filtering stage, its conversion gain, noise figure, linearity, and port-to-port isolation directly shape the overall receiver sensitivity and dynamic range.

The Gilbert-cell topology is the dominant choice for active down-conversion mixers, offering conversion gain, inherent double-balanced operation, and a well-established design methodology [1], [2]. This work presents a complete design and characterization study of a Gilbert-cell mixer for the 5 GHz WLAN band in the TSMC 0.18 μm RF CMOS process. The emphasis is on the design methodology and the trade-offs between conversion gain, noise figure, and linearity, verified through a full set of harmonic-balance simulations. The mixer is intended to be driven by an on-chip voltage-controlled oscillator (VCO) as its LO, forming the core of an integrated receiver front-end.

II. CIRCUIT ARCHITECTURE

The mixer employs the classical double-balanced Gilbert-cell architecture (Fig. 1), comprising three vertically stacked stages between the 1.8 V supply and ground.

Transconductance stage: A differential NMOS pair (M_5 , M_6 ; $W/L = 3\mu\text{m}/0.18\mu\text{m}$, 32 fingers, 96 μm total width each) converts the differential RF voltage at 5.5 GHz into a signal current. As the input stage, it sets the conversion gain and dominates the noise and linearity of the mixer.

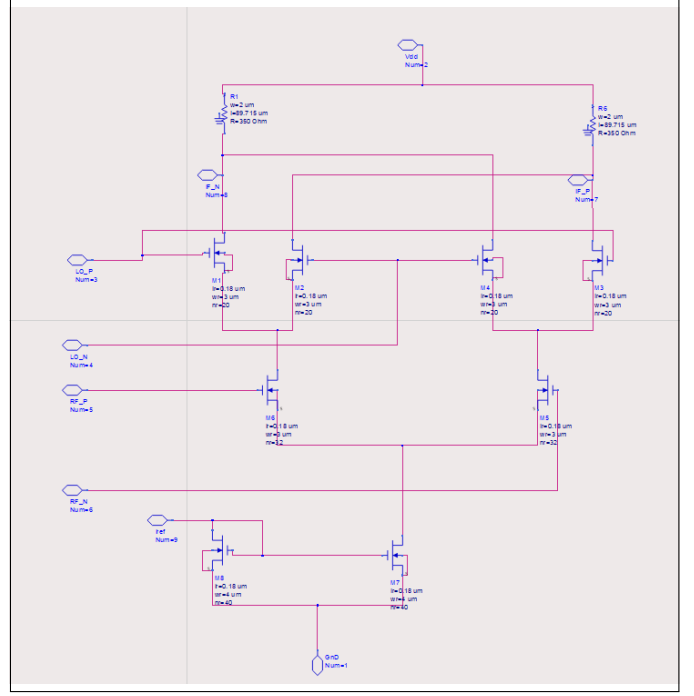


Fig. 1. Schematic of the double-balanced Gilbert-cell down-conversion mixer, showing the transconductance pair, the LO-driven switching quad, and the resistive IF loads.

Switching quad: Four NMOS devices (M_1 – M_4 ; $W/L = 3\mu\text{m}/0.18\mu\text{m}$, 20 fingers, 60 μm total each) driven by the differential LO commutate the RF signal current at the LO rate, performing the frequency translation.

Load stage: Two resistive loads ($R_L = 350\Omega$, P+ poly with silicide) convert the down-converted current at the IF nodes into a differential voltage at 200 MHz.

Bias: An NMOS current mirror (M_7 , M_8 ; $W/L = 4\mu\text{m}/0.18\mu\text{m}$, 40 fingers, 160 μm total each) sets the mixer core tail current (2 mA) from an external reference.

III. CIRCUIT ANALYSIS

A. Frequency Conversion and the Switching Function

The switching quad, driven by the LO, alternately routes the transconductor current to the two IF output nodes. Ideally the quad acts as a polarity-reversing switch, multiplying the

RF current by a square wave $s(t)$ of period $1/f_{LO}$, whose Fourier expansion is

$$s(t) = \frac{4}{\pi} \sum_{n=1,3,5,\dots}^{\infty} \frac{1}{n} \sin(2\pi n f_{LO} t). \quad (1)$$

Multiplying the RF current $i_{RF}(t) = g_m V_{RF} \cos(2\pi f_{RF} t)$ by $s(t)$ and retaining the fundamental ($n = 1$) term produces components at $f_{RF} \pm f_{LO}$. The wanted IF at $f_{IF} = f_{RF} - f_{LO}$ therefore carries a current amplitude of $(2/\pi) g_m V_{RF}$, from which the down-conversion efficiency of the ideal switch, $2/\pi$, directly follows.

B. Conversion Gain

Combining the switching efficiency with the resistive load, the voltage conversion gain is

$$A_{conv} \approx \frac{2}{\pi} g_m R_L. \quad (2)$$

With the transconductance of a MOSFET in saturation given by

$$g_m = \sqrt{2 \mu_n C_{ox} \frac{W}{L} I_D}, \quad (3)$$

the gain scales as $\sqrt{I_D}$ and $\sqrt{W/L}$ through g_m , and linearly with R_L . These same quantities set the noise, the linearity, and the available headroom, making conversion gain the central node of the design trade-off space.

C. Noise Figure

For a down-conversion mixer, noise from both the signal band and the image band folds onto the IF, so the single-sideband (SSB) noise figure is approximately 3 dB higher than the double-sideband value. The output noise is dominated by the input transconductor and the load resistors. The thermal noise current of the transconductance devices is

$$\overline{i_{n,d}^2} = 4kT \gamma g_m \Delta f, \quad (4)$$

and that of each load resistor is $\overline{i_{n,R}^2} = (4kT/R_L) \Delta f$. Referred to the mixer input, the load-resistor contribution is divided by the conversion gain; increasing g_m therefore both raises the gain and lowers the input-referred noise figure. Noise-contributor analysis of this design confirms that the IF load resistors dominate, indicating that active (current-source) loads would further reduce the noise figure.

D. Linearity and IIP3

The nonlinearity of the transconductor generates third-order intermodulation (IM3) products that fall in-band. Expanding the drain current in a power series about the bias point,

$$i_D = g_m v_{gs} + g_2 v_{gs}^2 + g_3 v_{gs}^3 + \dots, \quad (5)$$

the third-order coefficient g_3 governs IM3 generation. Under a two-tone excitation of amplitude A per tone, the fundamental output grows linearly ($\propto A$) while the IM3 output grows

as A^3 ; the extrapolated intercept at which the two are equal defines the input-referred IIP3,

$$V_{IIP3} = \sqrt{\frac{4}{3} \left| \frac{g_m}{g_3} \right|}, \quad (6)$$

with the practical extraction from simulated tone powers given by

$$IIP3 = P_{in} + \frac{P_{fund} - P_{IM3}}{2}. \quad (7)$$

Since IIP3 is set by the transconductor device physics through the ratio g_m/g_3 , it is largely independent of the load resistance and the bias current, and a fundamental trade-off exists between conversion gain and IIP3.

E. Port Isolation

In a fully balanced double-balanced cell, the LO and RF components appear as common-mode signals at the differential IF output and cancel:

$$V_{IF} = V_{IFP} - V_{IFN}, \quad (8)$$

so any component common to both nodes is rejected. Ideal isolation is therefore limited only by device and layout mismatch; at the schematic level, with perfect symmetry, LO-to-IF and RF-to-IF feedthrough are suppressed to the numerical noise floor.

IV. SIMULATION SETUP

The mixer was characterized in Keysight ADS using harmonic-balance (HB) analysis. The core was biased at a 2 mA tail current from the 1.8 V supply. The RF and LO were applied as differential tone pairs: the RF ports at 5.5 GHz with a 1.2 V common-mode gate bias, and the LO ports at 5.3 GHz with a 1.1 V common-mode gate bias, producing a down-converted IF at 200 MHz. The LO common-mode bias positions the switching-quad gates at their commutation point so that the applied LO drive fully steers the transconductor current; an LO-drive sweep confirmed that the conversion gain saturates once the quad switches completely.

Conversion gain and the 1-dB compression point were extracted from a single-tone RF power sweep (HB fundamentals: LO and RF, order 5 each). The noise figure was obtained from HB noise analysis at a small-signal RF input. IIP3 was extracted from a two-tone test using a three-fundamental HB simulation (LO, $f_1 = 5.499$ GHz and $f_2 = 5.501$ GHz), giving IF fundamentals at 199 and 201 MHz and IM3 products at 197 and 203 MHz. The IM3 products appear as the outer tones of the down-converted cluster and were read in the linear region, where the IIP3 characteristic of (7) is independent of input power.

V. SIMULATION RESULTS

A. Conversion Gain and Compression

Fig. 2 shows the conversion gain and IF output power versus RF input power. In the small-signal region the mixer provides a flat conversion gain of 17.9 dB. The gain compresses at higher input levels, giving an input-referred 1-dB compression point of -19 dBm.

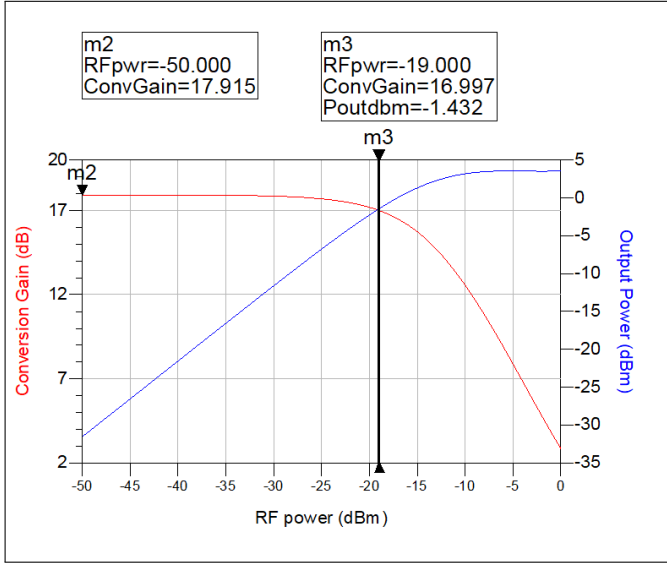


Fig. 2. Conversion gain and IF output power versus RF input power, showing a flat small-signal gain of 17.9 dB and an input-referred 1-dB compression point of -19 dBm.

B. Noise Figure

HB noise analysis at a small-signal RF input yields a single-sideband noise figure of 11.4 dB at the 200 MHz IF, consistent with the analysis of Section III-C.

C. Linearity

Fig. 3 shows the fundamental IF tone and the IM3 product versus input power, exhibiting the characteristic slope-one and slope-three behaviour predicted in Section III-D. Extrapolation in the linear region yields an input-referred IIP3 of -9.9 dBm, consistent with the measured P_{1dB} . This value is representative of a standard (nonlinearized) active Gilbert cell; linearization techniques such as multiple-gated transistors or derivative superposition offer a path to higher IIP3 in future work.

D. Output Spectrum and Isolation

Fig. 4 shows the differential IF output spectrum. The wanted IF tone at 200 MHz dominates, while the RF (5.5 GHz) and LO (5.3 GHz) feedthrough components are deeply suppressed, confirming the double-balanced cancellation of Section III-E. Higher-order mixing products fall well outside the IF band and are readily removed by IF filtering.

E. Performance Summary

Table I summarizes the simulated performance and compares it with reported 180 nm CMOS Gilbert-cell mixers.

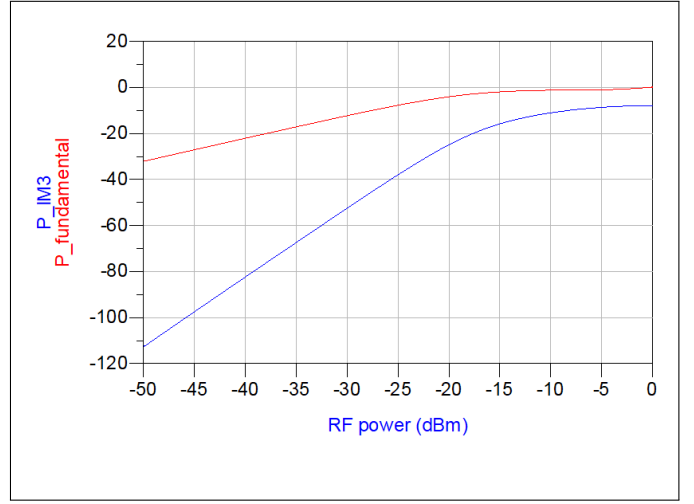


Fig. 3. Fundamental and third-order intermodulation output power versus input power, exhibiting the expected slope-one and slope-three behaviour. Extrapolation yields an input-referred IIP3 of -9.9 dBm.

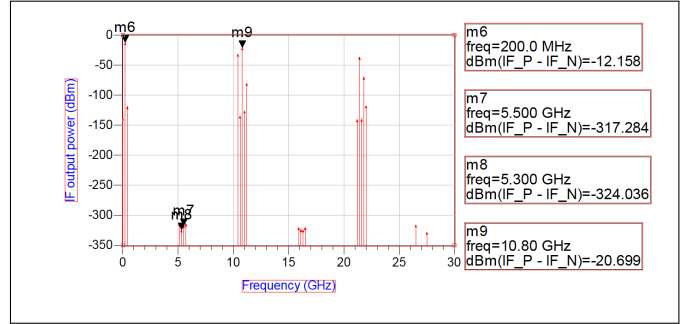


Fig. 4. Differential IF output spectrum, showing the wanted 200 MHz tone and the deeply suppressed LO and RF feedthrough.

TABLE I
PERFORMANCE SUMMARY AND COMPARISON

Parameter	This Work	[4]	[5]	[6]	[7]
Technology (nm)	180 CMOS	180 CMOS	180 CMOS	130 BiCMOS	130 CMOS
Supply (V)	1.8	1.8	1.8	1.8	1.5
RF (GHz)	5.5	2.4	1-10	13-23	23-25
IF (MHz)	200	10	100	100	100
Conv. gain (dB)	17.9	4.13	9.95	7.2	26.1
NF, SSB (dB)	11.4	14.77	8.12	8.5-12	7.7
P_{1dB} (dBm)	-19	-	-	-13	-17.8
IIP3 (dBm)	-9.9	11.5	-	-	-

VI. CONCLUSION

A double-balanced Gilbert-cell down-conversion mixer for the 5 GHz WLAN band has been designed and simulated in TSMC 0.18 μ m RF CMOS. The mixer achieves 17.9 dB conversion gain, an 11.4 dB SSB noise figure, a -19 dBm input P_{1dB} , and a -9.9 dBm IIP3 from a 1.8 V supply. The design serves as the down-conversion core of an integrated receiver front-end driven by an on-chip VCO. Future work includes replacing the resistive loads with active loads to lower the noise figure and applying transconductor linearization to improve IIP3.

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